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Universidad de [University Name]

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Master's Degree in [Name of the Master's Programme]

MASTER'S THESIS | Trabajo Fin de Máster

Title of Your Master's Thesis, Exactly as in the Approved Proposal

An optional subtitle that adds context

[Title of the thesis in the second language]

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ACADEMIC YEAR

2025/2026

[City], June 2026

Title of Your Master's Thesis, Exactly as in the Approved Proposal: An optional subtitle that adds context

Master's Thesis, Master's Degree in [Name of the Master's Programme], Universidad de [University Name], 2025/2026.

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Supervisor's Approval

Dr. First-name Surname and Dr. First-name Surname, supervisors of the Master's Thesis entitled *Title of Your Master's Thesis, Exactly as in the Approved Proposal*, carried out by **First-name Surname1 Surname2** within the Master's Degree in [Name of the Master's Programme], hereby authorise its submission for public defence and assessment at the [School or Faculty Name], Universidad de [University Name].

[City], June 2026

Dr. First-name Surname
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Dr. First-name Surname
Co-supervisor

Declaration of Originality

I, **First-name Surname1 Surname2**, student of the Master's Degree in [Name of the Master's Programme] at Universidad de [University Name], declare that:

1. This Master's Thesis, entitled *Title of Your Master's Thesis, Exactly as in the Approved Proposal*, is my own original work and has been carried out under the supervision of Dr. First-name Surname and Dr. First-name Surname.
2. It has not been submitted, in whole or in part, for any other degree or qualification at this or any other university.
3. All the sources used have been properly cited and acknowledged, and every figure, table or text taken from other works has been reproduced with permission or under the terms of its licence.
4. Any use of generative artificial intelligence tools has been declared in this document in accordance with the regulations of my university.
5. I am aware that plagiarism and any other form of academic dishonesty may lead to the failure of this Master's Thesis and to disciplinary measures.

[City], June 2026

First-name Surname1 Surname2

Author

To [whoever you want to dedicate this work to].

Acknowledgements

> **Template guidance**
>
> Thank the people and institutions that made this work possible: your supervisors, colleagues, family and friends. If the work was funded (a grant, a research project, a company internship) or used shared resources (a computing cluster, a dataset, a laboratory), acknowledge them here with the exact name and reference number that the funder requires. Keep it to one page.

I would like to thank my supervisors, Dr. First-name Surname and Dr. First-name Surname, for their guidance and support throughout this project. [Continue with your own acknowledgements.]

Abstract

Template guidance

The abstract is the most widely read part of your thesis, and often the only one that appears in your university's repository. Write it last, as a self-contained text of about 200–300 words (check the limit of your university) that covers, in this order: the *context* and the problem, the *objective*, the *methods*, the main *results* (with numbers when possible) and the *conclusions*. Avoid citations, undefined acronyms, equations and references to figures or tables.

This document is a LaTeX template for Master's theses written in English at Spanish public universities, and at the same time it is its own user guide. It provides a complete and configurable structure, from the cover page and the declarations commonly required by Spanish regulations to the chapters of a research-oriented thesis, the references and the appendices. All the personal and institutional data are defined in a single configuration file, so the same template can be used at any university by changing the university name, logos, colours and a few options. The template supports digital and printed versions, two typefaces, a second abstract in Spanish or in any co-official language, and guidance boxes that disappear automatically in the final version. It uses only the packages of the original template, so it is light and fast to compile. Replace this text with your own abstract.

Keywords: Master's thesis; LaTeX; template; Spanish universities; reproducibility

Resumen

[Title of the thesis in the second language]

> **Template guidance**
>
> Most Spanish universities require the title, the abstract and the keywords also in Spanish (or in the co-official language of the region) when the thesis is written in English. Write here a faithful translation of your English abstract. The text inside this environment is hyphenated according to the rules of that language. The title and the keywords come from `\ThesisTitleSecondLanguage` and `\KeywordsSecondLanguage` in `config.tex`.

[Write here the translation of your abstract into the second language.]

Palabras clave: [keyword 1; keyword 2; keyword 3]

Statement on the Use of Generative AI

Template guidance

Be specific: name each tool and its version, explain what you used it for and in which parts of the thesis, and state that you have reviewed the output and take full responsibility for the content. Some universities only allow AI for language editing, others require your supervisor's authorisation for other uses. If you did not use any generative AI tool, keep only the alternative sentence below.

In accordance with the regulations of Universidad de [University Name], I declare that I have used the generative artificial intelligence tools listed in Table 1, for the purposes and in the parts of this Master's Thesis indicated there.

Table 1. Generative AI tools used in the preparation of this thesis.

Tool (version)	Purpose	Parts affected
[Tool name] ([version])	Language editing of drafts written by the author	Chapters 1–3
[Tool name] ([version])	Help with debugging the analysis code	Code repository

I have critically reviewed and, where necessary, edited all the output of these tools. The research questions, methodology, analysis, interpretation of the results and conclusions are my own, and I take full responsibility for the content of this work.

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List of Acronyms

AI	Artificial intelligence
DOI	Digital Object Identifier
ECTS	European Credit Transfer and Accumulation System
GDPR	General Data Protection Regulation
GPU	Graphics processing unit
ML	Machine learning
SDG	Sustainable Development Goal
TFM	*Trabajo Fin de Máster* (Master's thesis)

CHAPTER 1

Introduction

> **Template guidance**
>
> The introduction is a high-level overview of the whole work. Explain the motivation and why the problem matters, how it has been approached so far (briefly; the details belong to Chapter 2), what is missing and what this thesis contributes, without going into technical detail. Finish with the structure of the document, one or two sentences per chapter. Write it, or at least revise it, at the end, when you know exactly what the thesis contains.

1.1 Motivation

[Describe the context of the problem and why it is relevant: scientific, social, clinical, economic or industrial impact. Support your claims with references.]

1.2 Problem statement

[State clearly and concisely the problem addressed in this thesis and the gap in the current knowledge or practice that it aims to fill.]

1.3 Contributions

[List the main contributions of the thesis. Be concrete: a new method, a dataset, an empirical comparison, a software tool, a validated prototype. . .]

1.4 Structure of the document

The rest of this document is organised as follows. Chapter 2 reviews the theoretical background and the state of the art. Chapter 3 defines the objectives of the thesis. Chapter 4 describes the materials and methods, and Chapter 5 presents the results, which are discussed in Chapter 6. Finally, Chapter 7 summarises the conclusions and outlines future work.

CHAPTER 2

Background and Related Work

> **Template guidance**
>
> Show that you know the field. First describe the problem and the concepts and techniques needed to understand the thesis (theoretical framework); then review critically the most relevant work published on the topic (state of the art), grouping it by approach rather than listing papers one by one. End by identifying the gap that your work addresses. References are essential here: search the scientific literature (Scopus, Web of Science, Google Scholar, PubMed, IEEE Xplore, ACM Digital Library…) and cite every source.

2.1 Theoretical framework

[Introduce the concepts, definitions and techniques on which the thesis is built. For example, an introduction to machine learning for multi-label problems can be found in [1].]

2.2 State of the art

[Review and compare the most relevant previous works. A summary table that compares the approaches along a few dimensions is often very useful.]

2.3 Summary and research gap

[Summarise what is known and, most importantly, what is not known yet or not solved, which motivates the objectives in Chapter 3.]

CHAPTER 3

Objectives

> **Template guidance**
>
> State the general objective of the thesis from a high-level perspective, and break it down into specific, measurable objectives. A reader should get a clear idea of what has been done just by reading this chapter. If your work is hypothesis-driven, state the hypothesis or research questions first. The specific objectives will be revisited in Chapter 7 to assess whether they have been achieved.

3.1 Research questions

RQ1. [First research question]

RQ2. [Second research question]

3.2 General objective

[One sentence describing the overall aim of the thesis.]

3.3 Specific objectives

O1. [Review the state of the art on . . .]

O2. [Collect and prepare the data . . .]

O3. [Design and implement . . .]

O4. [Evaluate . . . and compare with . . .]

3.4 Scope and limitations

[Define what is within the scope of the thesis and what is deliberately left out.]

CHAPTER 4

Materials and Methods

Template guidance

Describe, in enough detail for another person to reproduce your work, the *materials* (data, participants, equipment, software) and the *methods* (techniques, algorithms, experimental design, evaluation) used to achieve the objectives. Justify every choice. Explain how the data were obtained and prepared, which techniques were used and why, how the experiments were run (hardware and software configuration) and how the results were evaluated. If your work involves people or personal data, the ethical and legal section is mandatory.

4.1 Work plan

[Describe the methodology followed to organise the work and its time plan. Figure 4.1 shows a simple Gantt chart made with an ordinary table.]

Task	Weeks 1–24
Literature review	
Data collection and preparation	
Method design and implementation	
Experiments and evaluation	
Writing the thesis	

Figure 4.1. Example time plan of the thesis.

4.2 Data

[Source of the data, description (size, variables, classes, period), quality issues and pre-processing. A table describing the variables is usually helpful.]

4.3 Methods

[Techniques and algorithms used, with references and the justification of the choice.]

4.4 Experimental setup

[Hardware and software configuration, hyperparameters, random seeds and the validation strategy. Table 4.1 is an example of a hardware requirements table.]

Table 4.1. Hardware requirements of the computer on which the system will be installed.

Component	Specification	Description
Processor	x86-64 architecture	Latest-generation multi-core processor, able to run the containers that make up the system in parallel
RAM	64 GB	Enough memory is essential for the system to work correctly
Storage	10 TB	Mass storage for the images, videos, AI models and other elements of the system; at least 7200 rpm is recommended
GPU	CUDA CC $\geq$ 3.5	GPU with CUDA Compute Capability 3.5 or later and at least 8 GB of memory to run the AI models
Network	Gigabit Ethernet	Access to external data sources and content sharing

4.5 Evaluation

[Metrics, baselines, statistical tests and how the results are compared.]

4.6 Ethical and legal considerations

[If the work uses personal data, describe how the General Data Protection Regulation (GDPR) [2] and the Spanish Organic Law 3/2018 on the Protection of Personal Data and Guarantee of Digital Rights [3] have been complied with: anonymisation or pseudonymisation, legal basis, informed consent, storage and access. Mention the approval of the ethics committee, if needed, with its reference number.]

4.7 Reproducibility

[State where the code and, if possible, the data can be found (a public repository with a DOI, e.g. on Zenodo), the software versions used [4] and any restriction on data access.]

CHAPTER 5

Results

> **Template guidance**
>
> Report all the results obtained by applying the methods to the materials, usually in tables and figures, in the same order as the experiments were described in Chapter 4. Describe what the results show objectively, without interpreting them yet: the interpretation belongs to Chapter 6 (some theses merge both chapters into a single 'Results and Discussion' chapter). Whenever possible, compare different methods and include measures of dispersion and statistical significance.

5.1 [First experiment]

[Results of the first experiment. Table 5.1 shows an example of a results table, in which the best result of each column is highlighted in bold.]

Table 5.1. Example of a results table: mean ± standard deviation over 10 runs. The best value of each column is shown in bold.

Method	Accuracy	F1 score	Time (s)
Baseline	0.812 ± 0.014	0.774 ± 0.019	12.4
Method A	0.845 ± 0.011	0.801 ± 0.016	48.9
Method B	0.851 ± 0.009	0.815 ± 0.012	103.2
Proposed method	**0.873 ± 0.008**	**0.842 ± 0.010**	61.7

5.2 [Second experiment]

[Results of the second experiment.]

5.3 Comparison with the state of the art

[Compare your results with those published in the literature reviewed in Chapter 2.]

CHAPTER 6

Discussion

> **Template guidance**
>
> Interpret the results: what do they mean, how do they answer the research questions, how do they compare with previous work and why. Be honest about the limitations of the study and about any threats to the validity of the conclusions. Discuss the practical implications of the findings.

6.1 Interpretation of the results

[Explain the results of Chapter 5 and answer RQ1 and RQ2.]

6.2 Comparison with previous work

[Relate your findings to the literature.]

6.3 Limitations

[Limitations of the data, the methods and the evaluation.]

6.4 Implications

[Practical, scientific, social or clinical implications of the work.]

CHAPTER 7

Conclusions and Future Work

> **Template guidance**
>
> Summarise the work done and the main results, and state clearly whether the objectives set in Chapter 3 (those of the thesis proposal approved by the academic committee) have been achieved. Do not introduce new results here. Finish with the lines of future work that derive from the thesis. Some universities also ask for a personal assessment of the competences acquired.

7.1 Conclusions

[Main conclusions of the thesis.]

7.2 Fulfilment of the objectives

Table 7.1 relates each specific objective to the part of the thesis where it has been addressed.

Table 7.1. Fulfilment of the specific objectives of the thesis.

Objective	How it has been achieved	Where
O1	[Brief description]	Chapter 2
O2	[Brief description]	Section 4.2
O3	[Brief description]	Section 4.3
O4	[Brief description]	Chapter 5

7.3 Future work

[Possible improvements and new lines of research that derive from this work.]

References

[1] Francisco Herrera, Francisco Charte, Antonio J. Rivera, and María J. del Jesus. *Multilabel Classification: Problem Analysis, Metrics and Techniques*. Springer, Cham, 2016. ISBN 978-3-319-41110-1. doi: 10.1007/978-3-319-41111-8.

[2] European Parliament and Council of the European Union. Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L 119, pp. 1–88, May 2016. URL https://eur-lex.europa.eu/eli/reg/2016/679/oj.

[3] Jefatura del Estado. Ley Orgánica 3/2018, de 5 de diciembre, de Protección de Datos Personales y garantía de los derechos digitales. Boletín Oficial del Estado, no. 294, December 2018. URL https://www.boe.es/eli/es/lo/2018/12/05/3.

[4] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

APPENDIX A

Alignment with the Sustainable Development Goals

Template guidance

Several Spanish universities ask students to reflect on how their thesis relates to the Sustainable Development Goals (SDGs) of the United Nations 2030 Agenda. Mark the degree of relationship of your work with each goal by changing the last argument of each `\sdg` command to `H` (high), `M` (medium), `L` (low) or `N` (not applicable), and then justify the most relevant ones. If your university does not ask for it, remove this appendix from `main.tex`.

Table A.1. Degree of relationship of this thesis with the Sustainable Development Goals.

Sustainable Development Goal	High	Medium	Low	N/A
SDG 1. No poverty				•
SDG 2. Zero hunger				•
SDG 3. Good health and well-being	•			
SDG 4. Quality education			•	
SDG 5. Gender equality				•
SDG 6. Clean water and sanitation				•
SDG 7. Affordable and clean energy				•
SDG 8. Decent work and economic growth				•
SDG 9. Industry, innovation and infrastructure		•		
SDG 10. Reduced inequalities			•	
SDG 11. Sustainable cities and communities				•
SDG 12. Responsible consumption and production				•
SDG 13. Climate action				•
SDG 14. Life below water				•
SDG 15. Life on land				•
SDG 16. Peace, justice and strong institutions				•
SDG 17. Partnerships for the goals				•

[Explain how the thesis contributes to the goals marked as high or medium, and its possible negative impacts, if any.]

APPENDIX B

Supplementary Material

Template guidance

Appendices contain material that supports the thesis but would interrupt the reading of the main text: additional tables and figures, questionnaires, consent forms, detailed proofs, user manuals, extended code listings, etc. Appendices are numbered with letters and referenced from the text like chapters, e.g. Appendix A. Single pages of a PDF document can be inserted as figures with `\includegraphics[page=2, width=\textwidth]{file.pdf}`.

B.1 Source code

[Link to the repository where the code of this thesis is available, e.g. https://github.com/username/repository, and to its archived version with a DOI.]

B.2 Additional results

[Additional tables and figures.]